



The Great Sorting Challenge

Organizing Our Messy World

Grade 6 | Computer Science

Planrs Academy Master Series

The Messy Library

Imagine walking into a library where books are scattered everywhere—on tables, floors, and random shelves!

You need to find "**Wings of Fire**" by APJ Abdul Kalam for your school project. How do you find it quickly?

Searching randomly takes forever and is very frustrating. We need **Order**. Sorting helps us find things fast!



What is Sorting?

Sorting means arranging things in a specific order—like lining up from shortest to tallest!

Indian Context Example:

Think of your **Morning Assembly**. Everyone stands in a neat line based on their height. That is sorting in action!

Sorting helps us organize and find things quickly and easily in our daily lives.

Bubble Sort

The Neighbor Swap

Bubble Sort is a simple way to sort by comparing items right next to each other and swapping them if they are in the wrong order.

Think about it: Imagine sorting cricket bats by their length. You compare two at a time and swap them if the first one is longer than the second!



How to Bubble Sort

Follow these simple steps to sort cricket bats from shortest to longest:

- 1 Look at two cricket bats next to each other.
- 2 If the first is longer than the second, swap them!
- 3 Move to the next pair and repeat the same check.
- 4 Keep going until you can go through the whole line without needing any swaps. Done!

Selection Sort

The Smallest First

Selection Sort means finding the smallest item in the whole group and putting it at the very start.

Imagine: You are picking mangoes from a basket. You always look through the whole basket to find the **smallest** mango first and move it to a new box!



How to Selection Sort

Follow these steps to sort mangoes into a new box, smallest first:

- 1 Look through the whole basket to find the smallest mango.
- 2 Put that mango in the new box at the front.
- 3 Repeat the process with the rest of the mangoes in the basket.
- 4 Continue until all mangoes are moved in perfect order!

Merge Sort

The Team Split

Merge Sort uses a clever trick called **Divide and Conquer**. It breaks a big group into smaller groups, sorts each one, and then joins them back together.

Example: Imagine two friends sort their own small card collections separately. Then, they merge their sorted cards into one big, perfectly ordered deck!



How Merge Sort Works

A simple three-step process to organize items using a deck of cards:

SPLIT: Divide the deck of cards into two equal halves.

SORT: Arrange each half in order separately (you can even give one half to a friend!).

MERGE: Combine both sorted halves carefully into one big sorted deck.

Divide and Conquer makes big problems much easier to solve!

Which Method Works Best?

Let's compare our three sorting algorithms to see when to use each one:

Bubble Sort

Easy neighbor swaps.

Best for: Small sets like a few cricket bats.

Selection Sort

Finds smallest each time.

Best for: Careful picking like choosing mangoes.

Merge Sort

Splits and joins.

Best for: Big sets like card collections or databases.

Think About It!

Which sorting method would you choose for these situations? Discuss with your classmates!

Situation A: A small box of 5 crayons.

Hint: Should you use easy neighbor swaps or a complex split?

Situation B: A big school database with 2,000 student names.

Hint: Is it better to split the work and merge it back?

Recap: What We Learned

- **Sorting** means arranging things in order (like height order).
- It helps us find things **quickly** in daily life.
- **Bubble Sort** swaps neighbors until everything is in place.
- **Selection Sort** searches for the smallest item first.
- **Merge Sort** splits, sorts, and joins back together!



Quiz Time!

Let's test what you've learned about sorting algorithms!

Q1: What does Bubble Sort do with two neighbors?

A) Ignores them

B) Swaps them if they are in the wrong order

C) Deletes one of them

Q2: In Selection Sort, what do you look for first?

A) The biggest item

B) A random item

C) The smallest item

Quiz Time!

Q3: What does Merge Sort do before joining groups?

A) It splits the group and sorts each half

B) It mixes the groups randomly

C) It hides the groups

Q4: Why do we sort things in real life?

Write your answer here...



Sorting Master!

Congratulations! You have completed The Great Sorting Challenge.

Now you can sort like a pro! Try sorting your books, toys, or crayons at home using these fun methods.

Keep Sorting & Stay Curious!